

Department of Electrical, Electronic and Computer Engineering

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Design and Implementation of a Secure Windows Server 2019 Infrastructure for Enterprise Network Services

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by

Group 1

Group project report submitted in partial fulfilment of the requirements for the degree

Bachelor of Engineering Technology in Computer Engineering

in the Faculty of Engineering and the Built Environment

at the Cape Peninsula University of Technology

Examiner: Mr. Vusumuzi Moyo
Course Code: OPS260S - OPERATING SYSTEMS 2

Bellville Campus
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Abstract

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Keywords: Operating Systems, Kernel Architecture, Resource Utilization, Performance Optimization, Memory Management, Linux, Windows

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2. DHCP Configuration

2.1 Background

Every machine, from computers to servers, requires an IP address for it to be able to send and receive packets of data or information. ([1]) These IP addresses act like home addresses on a street, and whilst DNS is responsible for resolving those IP addresses into names, there are two ways of assigning IP addresses to machines. These methods are static addressing and DHCP ([1]).

Static Addressing is the process of configuring IP addresses on a system manually. It is a hands-on approach where IP addresses are physically plugged into the NIC settings on a particular device ([1]). Despite its simplicity, this method is not scalable in environments that host multiple devices like a small or large company, where there are laptops, desktops, printers, servers or other types of machines ([1]). Each of these devices needs an IP address, however manually assign them becomes tedious at a larger scale. The solution to this problem lies in the Dynamic Host Configuration Protocol (DHCP) ([1]).

DHCP is a protocol that solves the problem of providing IP addresses for many devices on a wide scale. It allows for machines or devices that are plugged into the network and obtain IP address information ([1]). This ability to assign IP address information to devices allows for DHCP to have multiple applications. When a laptop or phone is connected to a Wi-Fi router to gain internet access, a DHCP server allows for that device to route traffic on the network by assigning IP addressing information ([1]). When a device connects to public Wi-Fi, its DHCP server runs on the router itself and in businesses where Windows Server serves as a datacenter, DHCP services are most often hosted on one or more servers across the network ([1]).

2.2 Configuration Steps

2.2.1 DHCP Post-Install configuration

Like other roles, the DHCP server needs to be provided a role during the installation phase. After the DHCP role has been added to the server, it isn't yet properly configured.

The DHCP server can give clients two kinds of access groups: DHCP Administrators and DHCP Users ([2]). DHCP Users have read-only access to the server, which allows them to view, but not to modify server data, including DHCP server configuration, registry keys, DHCP log files, and the database ([2]). They cannot create scopes, modify options values, create reservations or exclusion ranges, and more. DHCP Administrators are able to view and modify data at the DHCP server ([2]). They can create and delete scopes, add reservations, change option values, create superscopes or perform any other activity needed to adminster the DHCP server ([2]). Despite this overseering control, they do not have unlimited administrative rights,because they on have rights on the local computer only ([2]).

These two groups are local to the server as well, meaning they cannot authorize DHCP Servers. This can only be done by Domain Admins([2]). Authorizing the server is needed because DHCP communication typically occurs before any user or computer authentiation. DHCP protocols are based on IP protocols as well, therefore, an unknown DHCP server could provide invalid information to clients. Authorization prevents this ([3]).

The authorization of the DHCP server in the wizard required for users to be given credentials that they would need to access the server on a user level. So far the only credential set was `OPS\Administrator`. For any users who attempt to connect to the server in future, they must use this username.

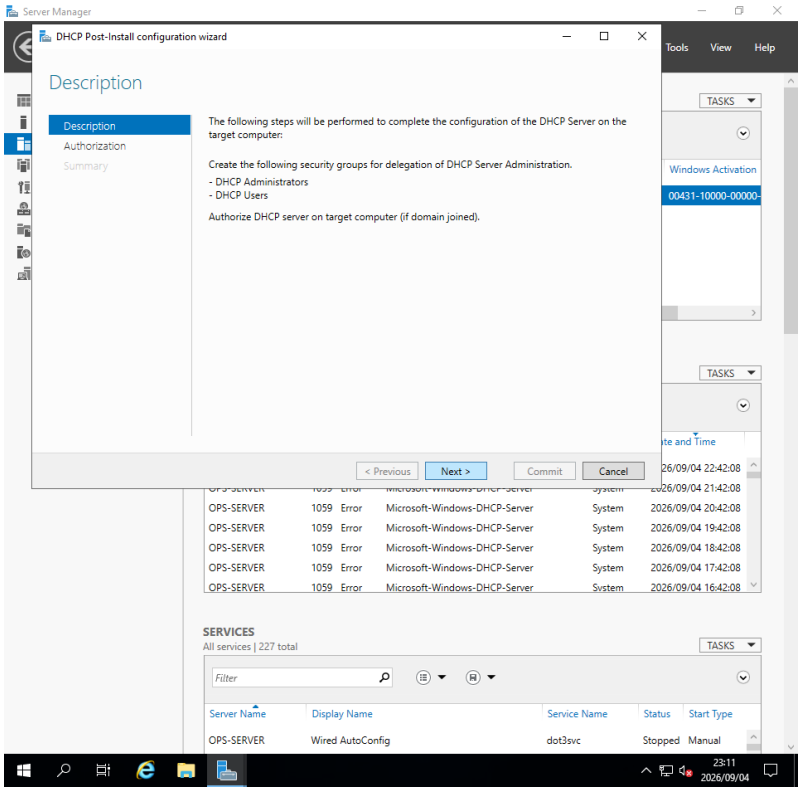


Figure 2.1: Description of DHCP Post-Install Configuration Wizard

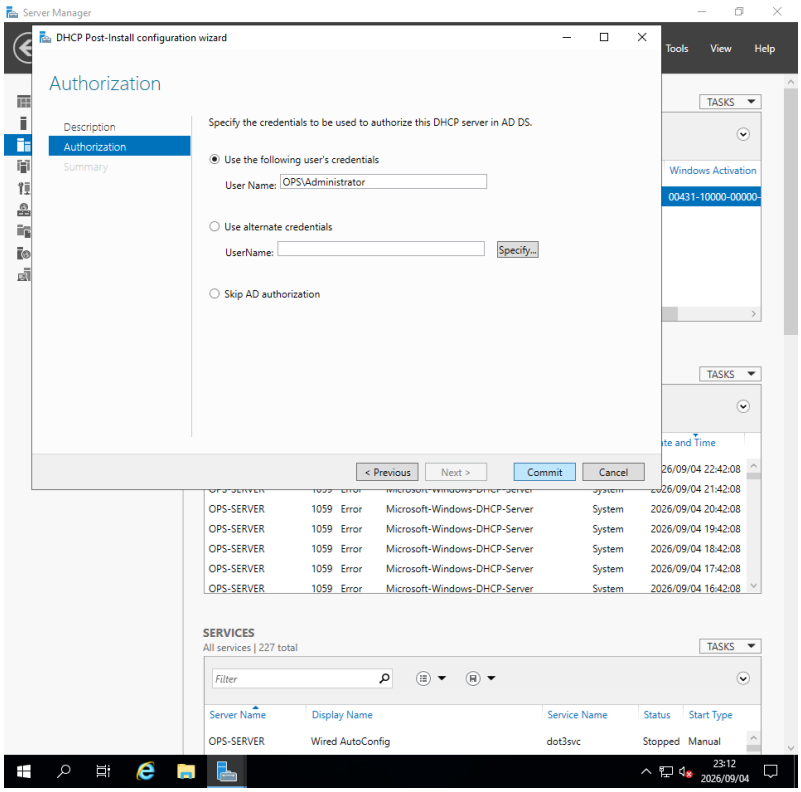


Figure 2.2: Authorization of DHCP Post-Install Configuration Wizard

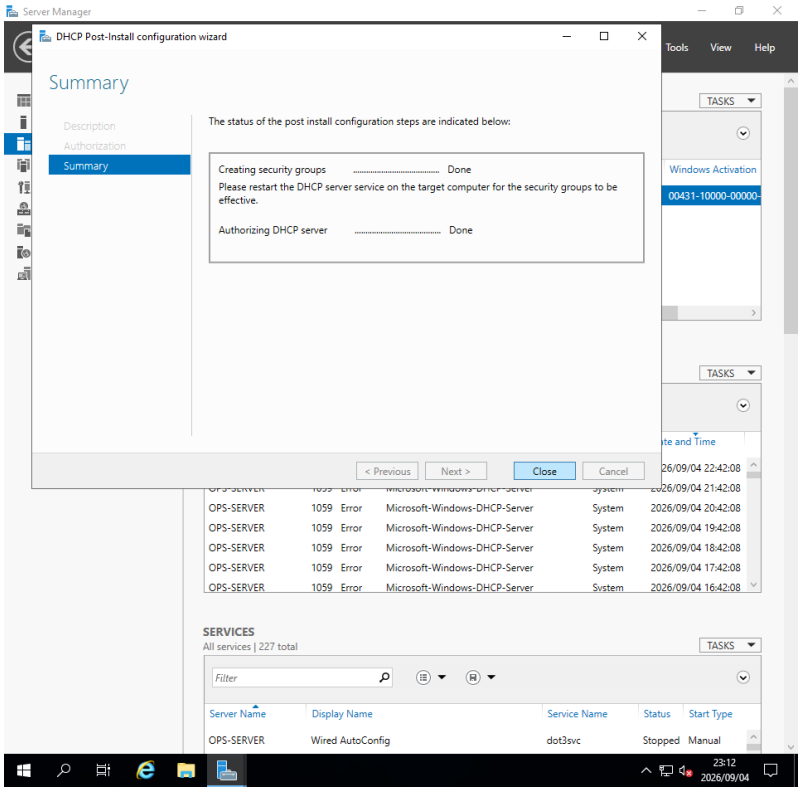


Figure 2.3: Post Install Configuration Completed

2.2.2 The DHCP Scope

For a DHCP server to hand out IP addresses, it needs to know IP addresses, subnet mask, default gateway and DNS server addresses are within the network so that it can package and hand out the information to computers who request it. This package of information is called a DHCP scope ([1]).

To begin defining the scope, we decided to launch a management tool using the Tools menu inside the Server Manager. This will open up a window for DHCP with the name of the device that the server is located on.

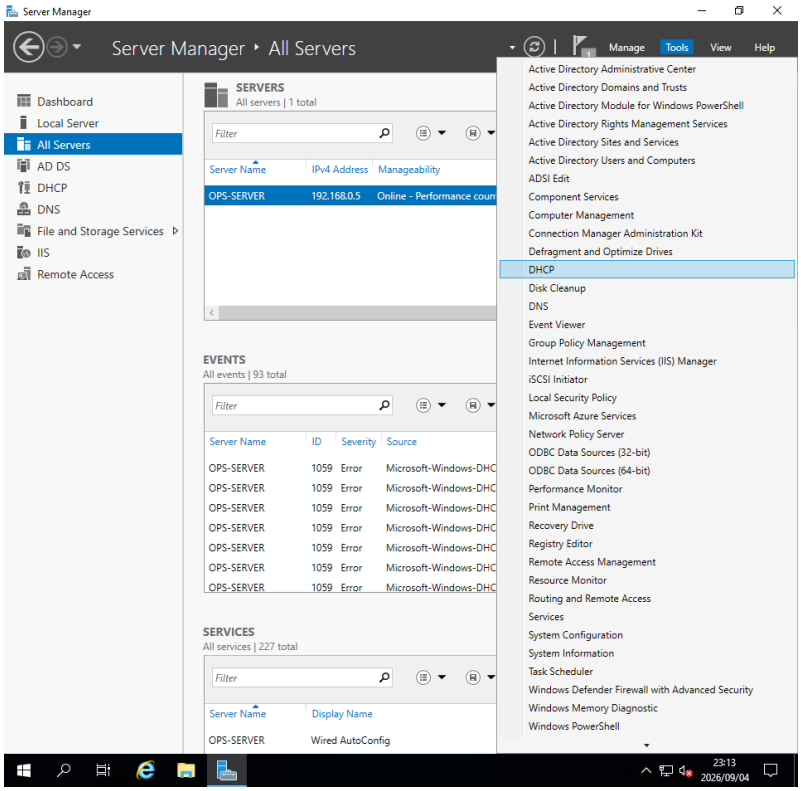


Figure 2.4: DHCP Selected on Tools Dropdown Menu

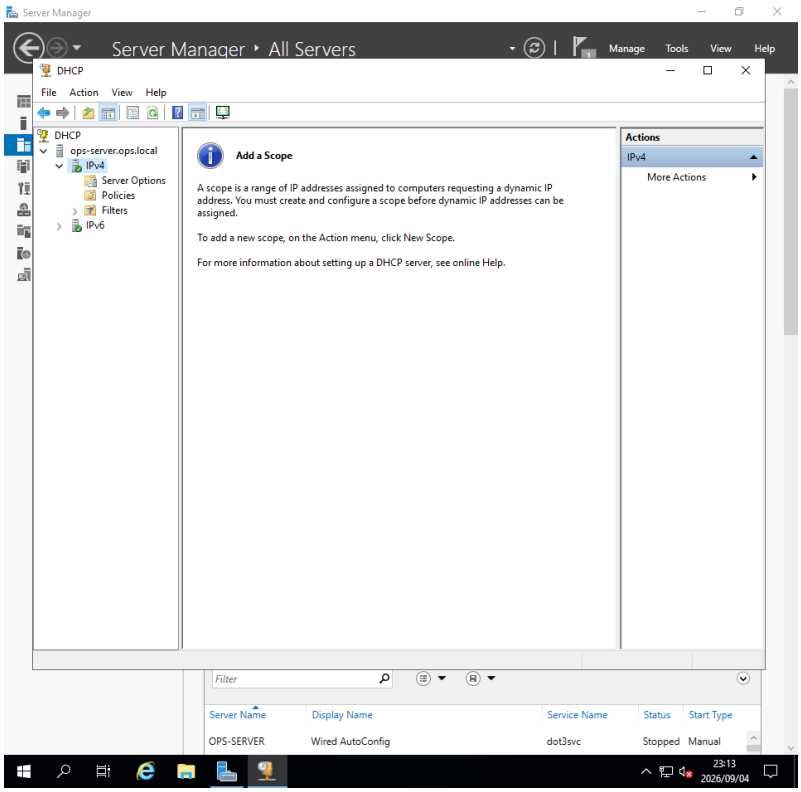


Figure 2.5: DHCP Windows

After opening the DHCP window, the computer was selected and dropped down two protocols: IPv4 and IPv6. The former was selected and first left-clicked on, where it gave information regarding the addition of a scope, along with the server options, policies and filters. This protocol did not have a scope as yet, so it needed to be configured. To properly begin the process, IPv4 was right-clicked on, giving a drop-down menu when the option to add a new scope was present. Choosing this option opened the New Scope Wizard. After clicking next on the wizard, the basics of the scope needed to be decided. The name given was GROUP 1-SCOPE.

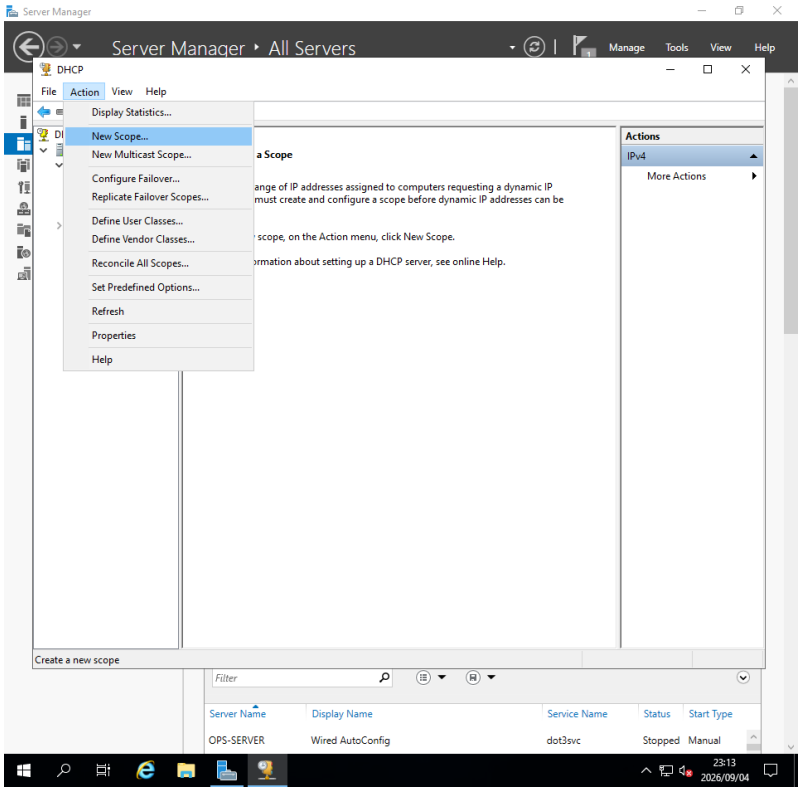


Figure 2.6: Adding a Scope

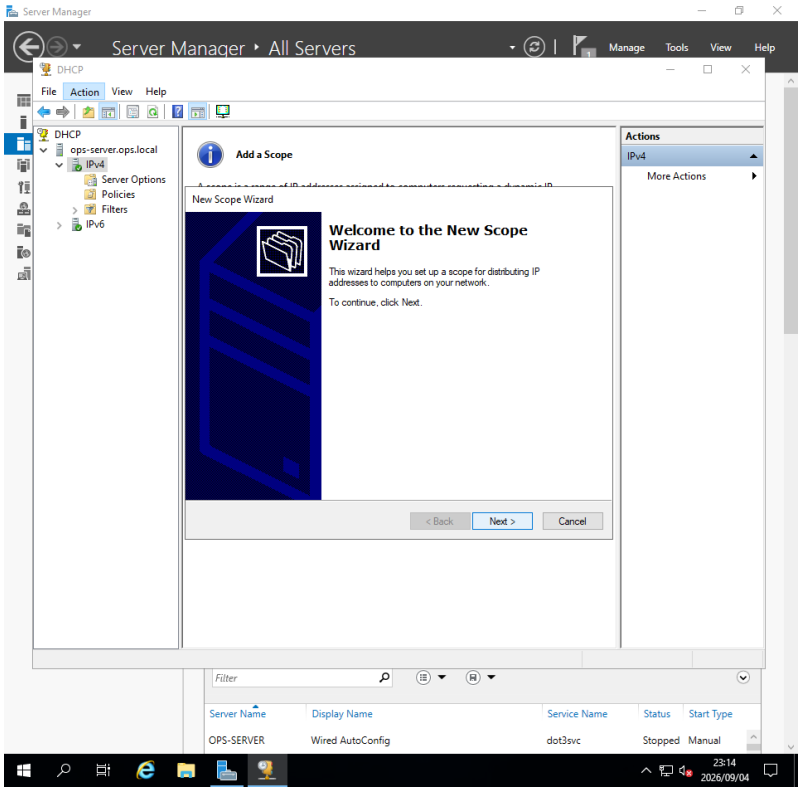


Figure 2.7: New Scope Wizard

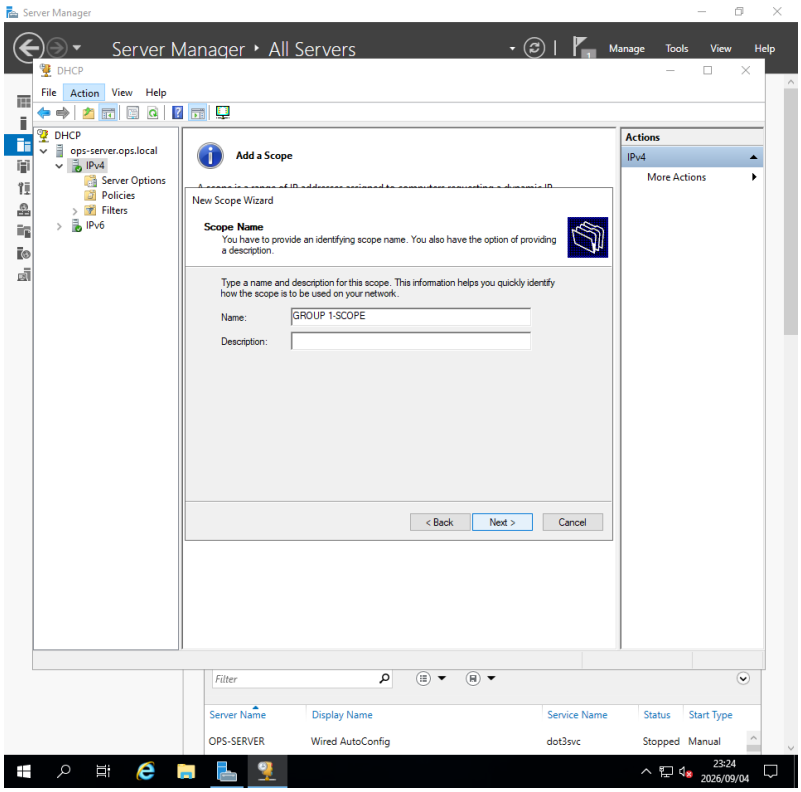


Figure 2.8: Group One Scope Made

There exists 4.3 billion posible IPv4 addresses and 3.7 billion available addresses, and therefore, there exists a countless amount of possible addresses that the scope can allow ([4]). A scope does not need to accomodate billions of addresses, it just needs a few hundred or a few dozen. When configuring it, the range of IP addresses needed to be decided. The addresses that the scope will hand out were from 192.168.0.100 to 192.168.0.254. This will be the address pool. There were no exclusions or delays made in the scope. Alongside handed out IPs, the scope also has the option to configure the amount of time that a client keeps an assigned IP before renewal, ensuring efficient management of IP addresses ([5]). This period of time is called the lease duration. It was left at the default number of 8 days.

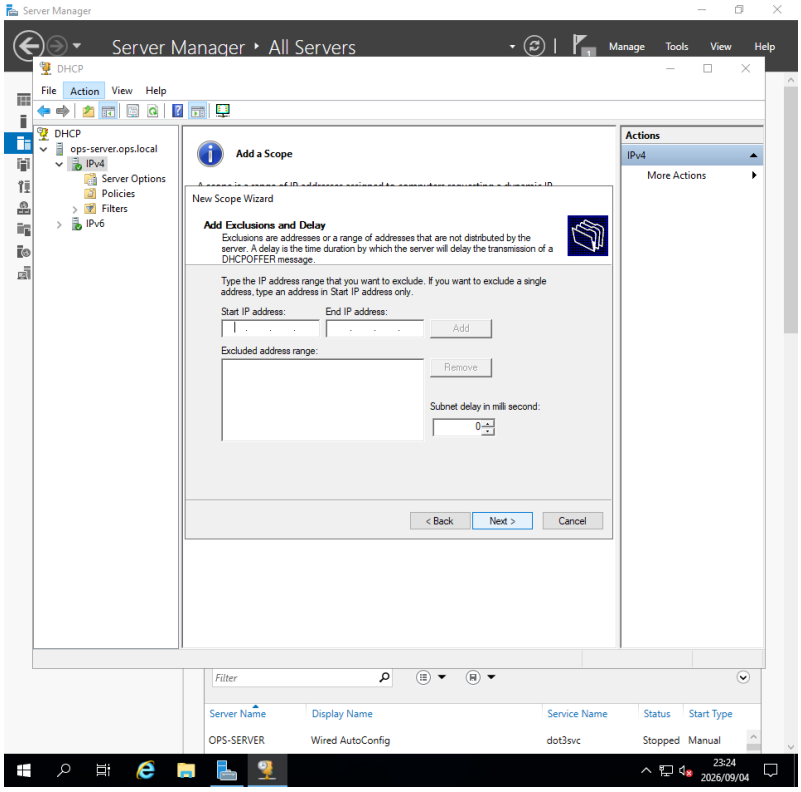


Figure 2.9: No Exclusions made

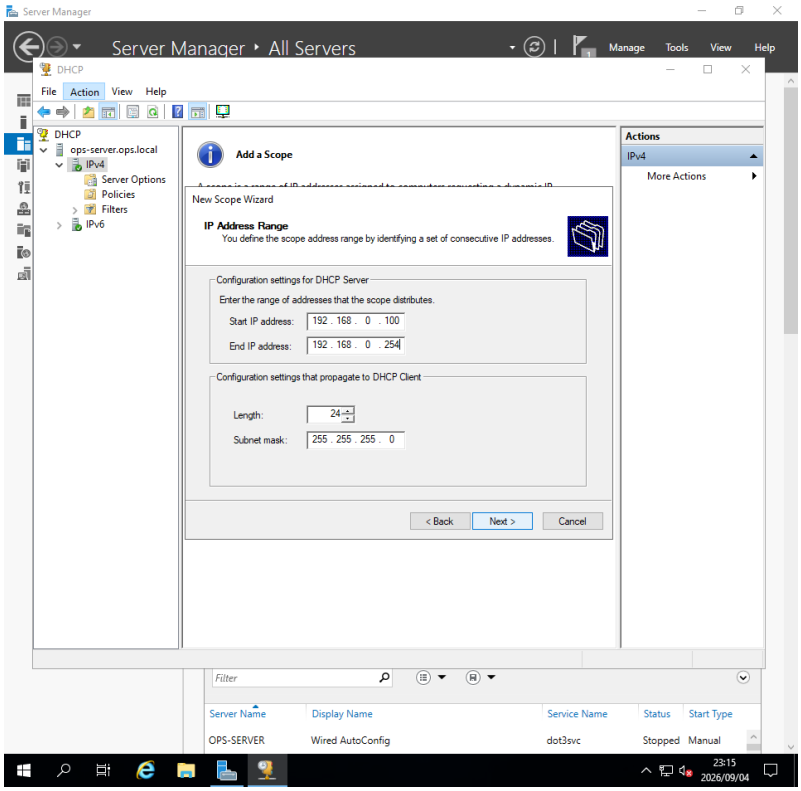


Figure 2.10: Setting IP Address Range

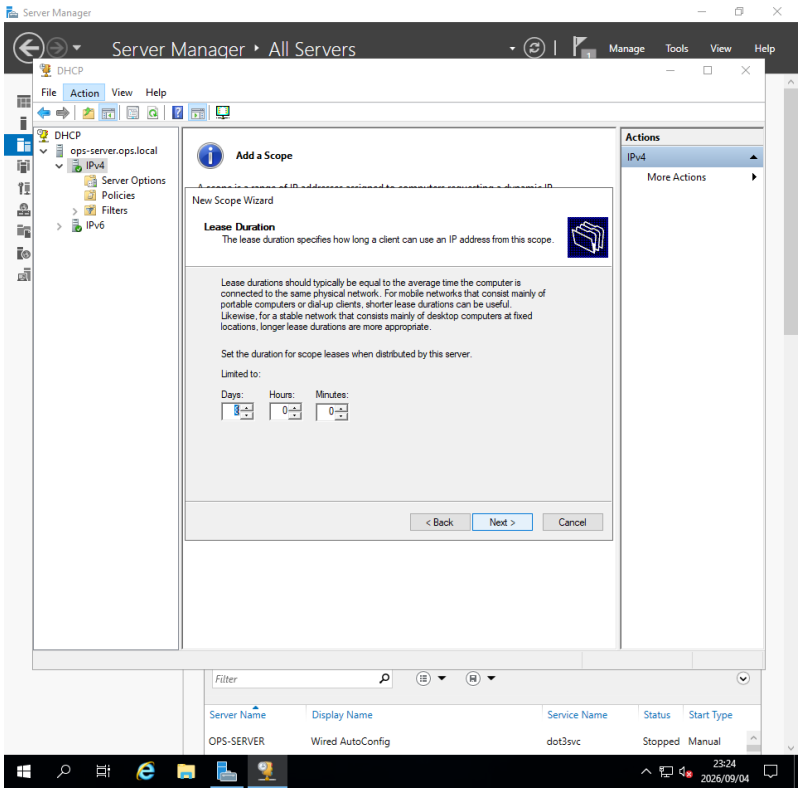


Figure 2.11: Lease duration of 8 days set

When clients use the server, access to the internet is an important feature that must be present in the server. In order for the clients to able to access the internet, they need to be able to talk to other devices and networks outside of the DHCP subnet. A default gateway allows for this. Without it, a client can only reach devices and networks inside its own subnet. The default gateway was given the IP address of 192.168.0.1.

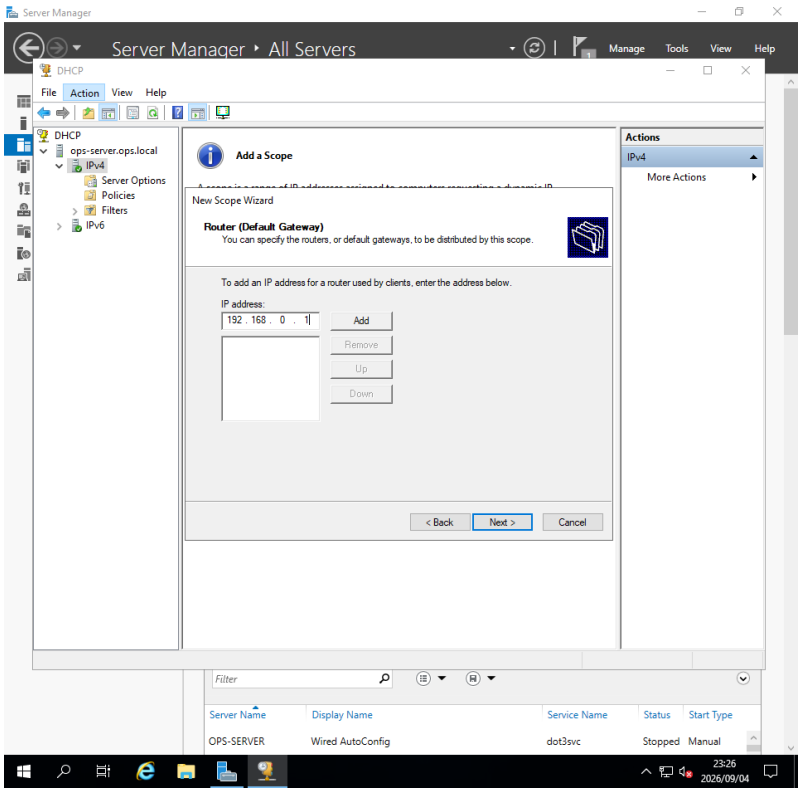


Figure 2.12: Default Gateway Configured (1)

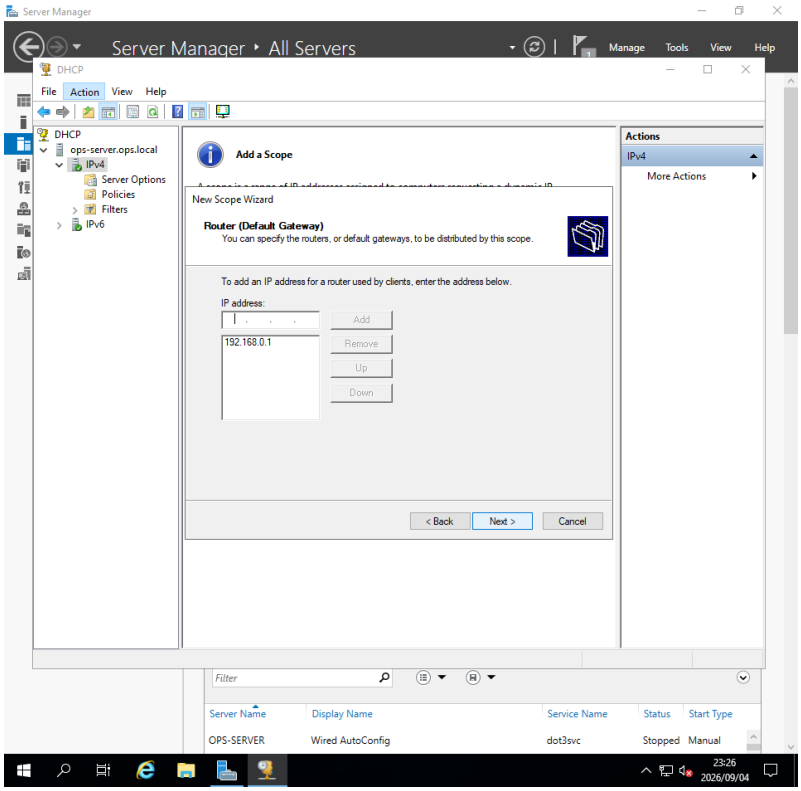


Figure 2.13: Default Gateway Configured (2)

After the setting up of the default gateway, the next step was setting up the DNS server. As mentioned earlier, the DHCP has the job of handing out IP addresses and more for a client to be able to use the network. An IP allows a device to communicate with other hosts, but IP addresses are inconvinient to memorize or type down. The DNS Server and Domain Name exist to allow for devices to access the DHCP server through use of IPs that resolve into domain names ([6]). The domain name was set to OPS.local and two IPs, 192.168.0.5 and 8.8.8.8, for two servers were set. This would mean that in order to access the server they would simply need to type in the domain name, making it more accessible for non-domain joined machines.

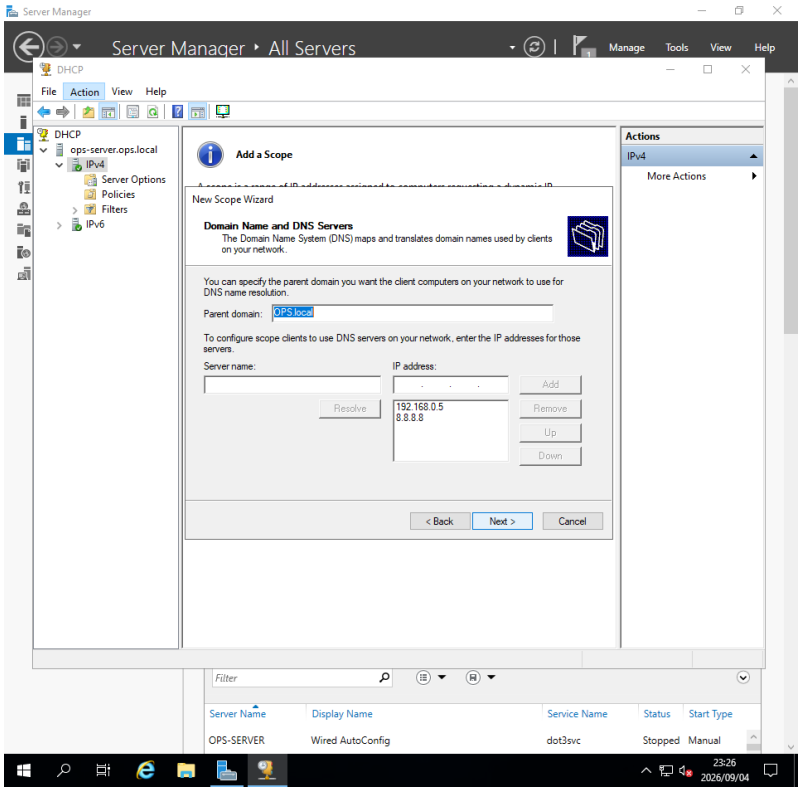


Figure 2.14: DNS Server and Domain Name

After the DNS Server and Domain Names are configured, there was an option for WINS Server configuration. WINS is a part of the scope that resolves NetBOIS computer names to IP Addresses. It is needed to support pre-Windows 2000 systems and older application using NetBOIS over TCP/IP ([7]). None of the systems planned to be connected in the network do not qualify as pre-Windows 2000 systems, there WINS is not needed.

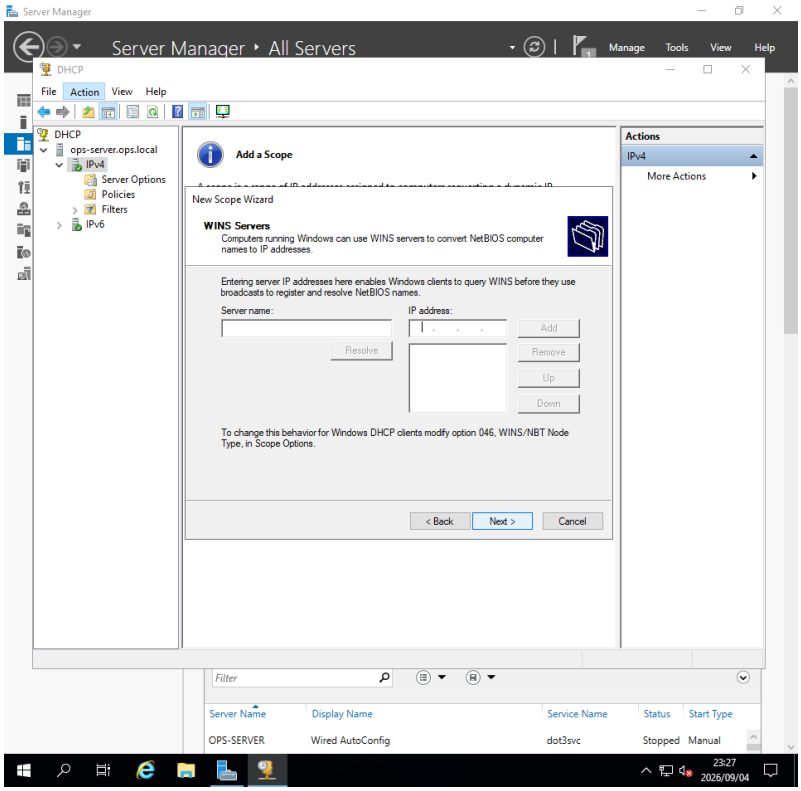


Figure 2.15: WINS Server configurinf left vacant

All necessary parts of the scope have been configured. After passing the WINS Server configuring, all that was left was activation of the scope. Upon activating the scope, the wizard will present completion display. To close the wizard, the finish button was clicked. The scope is now activated and ready for testing. The scope will now had out IP addresses from 192.168.0.100 to 192.168.0.254 and this will be its address pool. At the time of the creation of the scope, there were no address leases yet.

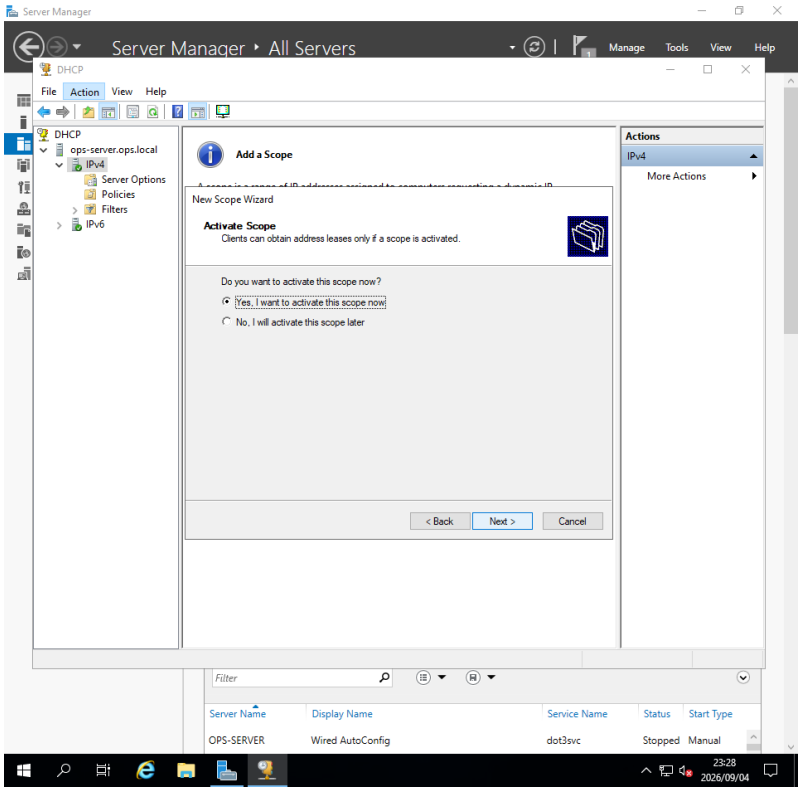


Figure 2.16: Activation of Scope

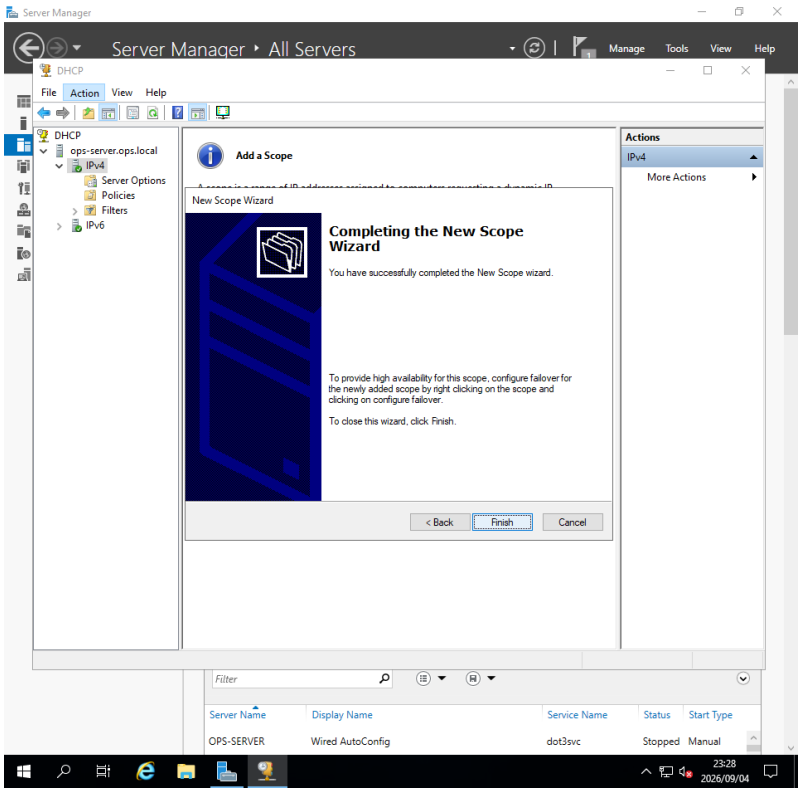


Figure 2.17: Finishing New Scope Wizard

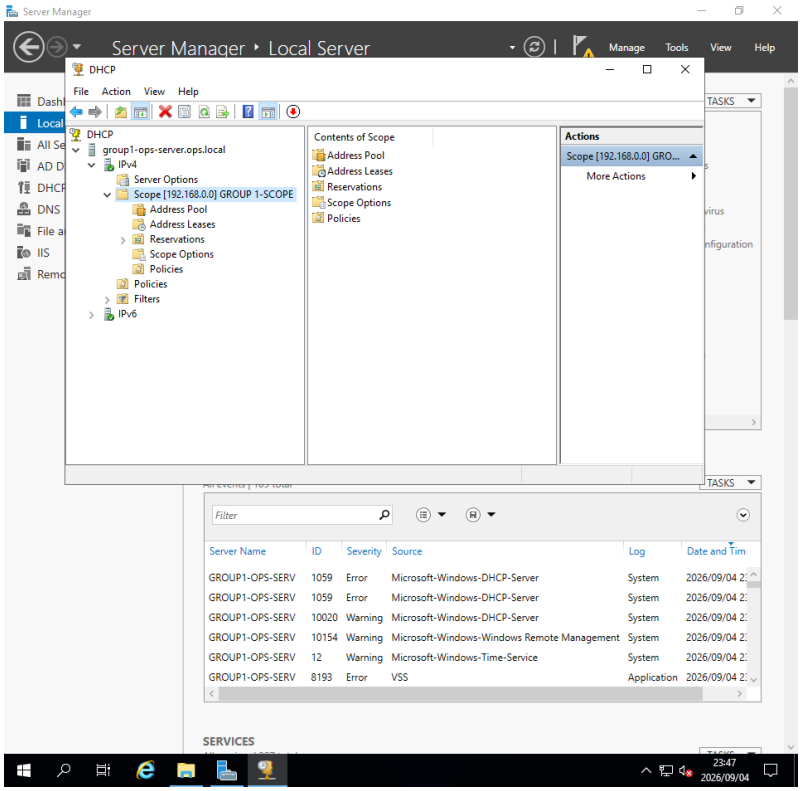


Figure 2.18: New DHCP Scope created

2.3 Issues encountered for DHCP Server Setup and Alternative Paths

2.3.1 Use of Powershell in event of DHCP Window Failure

On some occassions, the DHCP console can stop functioning due to either a domain name change or a new ethernet being added to it for internet access. This means that configuring a new scope using the window is not possible, however, this does not mean that new DHCP scopes cannot be configured or removed. Windows Powershell is still available on Windows Server 2019 and it serves as a way to bypass any surface level server issues. Using powershell commands allow you to still configure and modify DHCP scopes with a similar degree as if you were using the console. They can also be used when you simply prefer Powershell.

Some of the important commands you would need to configure DHCP IPv4 scopes with Powershell are as follows ([8]):

```
Add-DhcpServerv4Scope - adding scopes
Get-DhcpServerv4Scope - viewing scopes
Set-DhcpServerv4Scope - modify scopes
```

```
Remove-DhcpServerv4Scope - delete scopes
Get-DhcpServerv4ScopeStatistics - show usage statistics
Add-DhcpServerv4Lease - add lease
Get-DhcpServerv4Lease - view leases
Remove-DhcpServerv4Lease - remove Lease
Add-DhcpServerv4Reservation - add reservations
Get-DhcpServerv4Reservation - view reservations
Set-DhcpServerv4Reservation - modify reservations
Remove-DhcpServerv4Reservation - delete reservation
```

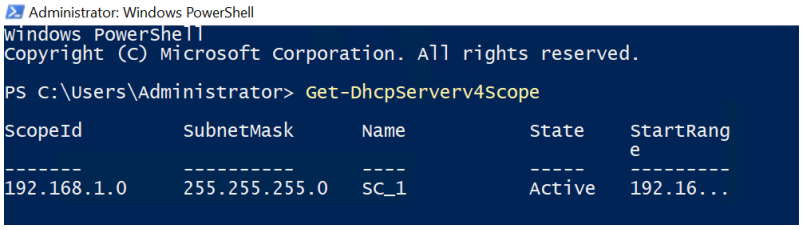


Figure 2.19: Example of accessing DHCP through Powershell

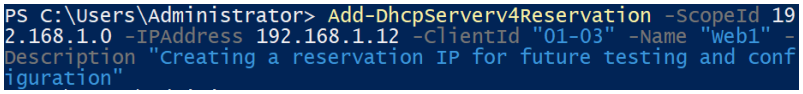


Figure 2.20: Example of adding a DHCP reservation



Figure 2.21: Example of verification of DHCP reservation

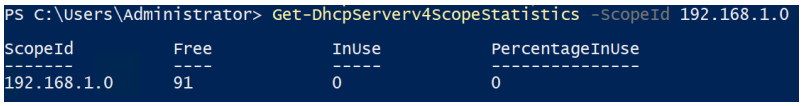


Figure 2.22: Example of IP address tracking

2.3.2 Use Windows Powershell to create DHCP scope

To create a DHCP scope using Windows Powershell, it must be opened as administrator([9]). Considering that scopes have most likely been made already, a command must be written to check for existing scopes.

The command Get-DhcpServerv4Scope will list every scope in the DHCP server that was created using the DHCP console ([9]). If there wasn't any preexisting scopes, nothing will show.

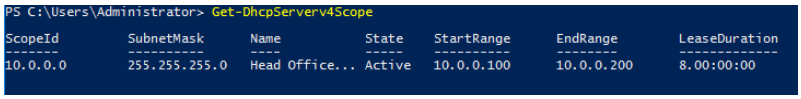


Figure 2.23: Example of Get-DhcpServerv4Scope command in use

To create scope in Powershell normally would involve a lot of typing, as the Add-DhcpServerv4Scope command has a lot of parameters. A conventional use of the full command would look like the following:

```
Add-DhcpServerv4Scope `
    -Name "Test Scope" `
    -Description "Scope for DHCP testing" `
    -StartRange 192.168.10.100 `
    -EndRange 192.168.10.200 `
    -SubnetMask 255.255.255.0 `
    -LeaseDuration 0.08:00:00 `
    -State Active `
    -PassThru
```

This can be tedious, so splatting can be used to make passing parameters easier([10]). Powershell Splatting is when you use a hashtable to create a parameter set that is passed to the cmdlet. The cmdlet will create DHCP scope according to the given parameters. For example, if you entered the following in the powershell console:

```
$HashArgs = @{
     'Name' = 'Head Office Wireless Scope';  # This sets the name of the wireless s
     'Description' = 'Wireless Clients';     # This is the human-readable descripti
     'StartRange' = '10.1.0.100';            # Specifies the starting IP address in
     'EndRange' = '10.1.0.200';              # Specifies the end IP address in the
     'SubnetMask' = '255.255.255.0';         # Specifies the subnet mask of the sco
     'State' = 'Active';                     # Activates the scope
     'LeaseDuration' = '1.00:00:00';         # Specifies the length of the lease du
 }
 Add-DhcpServerv4Scope @HashArgs
```

The hashtable @HashArgs is given DHCP server parameters When the Add-DhcpServerv4Scope command is entered, a new scope named Head Office Wireless Scope is made, with an address pool between 10.1.0.100 and 10.1.0.200, with a subnet of 255.255.255.0([9]). The lease duration is a day and the scope is made active. To check if the scope was made, run Get-DhcpServerv4Scope again. A way to make looking at the scopes easier is to add " | Format-List" at the end of the command([9]).

2.4 Testing

2.4.1 Checking Server

To test the operation of a DHCP server, there must be confirmation that data packets are travelling between the server and client devices. To start, we must check the server on a Powershell window that is run as administrator. We must run the command:

```
Get-Service -Name (Name of DHCP server)
```

The status collumn must display running. If the command shows the service as Stopped, that means that the server can't answer any client requests. The reason behind that issue, might be that the server somehow hasn't started or the server hasn't been authorized. If it needs to be started, run the command

```
Start-Service (Name of DHCP server)
```

after that running the get service command will help confirm if it running. Another probable reason is that the server needs authorization. The command

```
Get-DhcpServerInDC
```

will verify if the DHCP server is authorized in Active Directory([11]). However for this to work, you need to first add the server into the Directory with the command

```
Add-DhcpServerInDC -DnsName (name of server)
-IPAddress (address of server)
```

A few other checks to do are to check the configuration and the scope usage, seeing howing addresses are in use and how many are free, with the following commands([11]).

```
Get-DhcpServerv4ScopeStatistics
Get-DhcpServerv4Scope
```

2.4.2 Test from a client and Verifying on server

To test a server from a client, it will have to request a lease. To request a lease on a Windows client, you will have use Command Prompt and run ipconfig /release, then ipconfig /renew, then ipconfig /all. ipconfig /release releases a DHCP client lease, ipconfig /renew renews the client lease, ipconfig /all or just ipconfig verifies a DHCP client lease([12]).

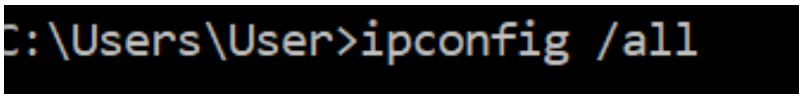


Figure 2.24: ipconfig /all command in Command prompt

The address, subnet mask, gateway, DNS servers and DHCP server line must all match your scope.

There is a chance that your client will have incorrect or missing options present. The cause of this could be that the network configuration isn't working correctly. This is due to the server clients not being able to support the missing options, but it also could be possible that the server isn't configured to hand them out ([13]).

The solution to this is to change the IP address list for the missing or incorrect options at the DHCP scope in use. You might have to configure the options as a Server Option at the DHCP server, remove it and set the correct value in the Scope Options for the DHCP scope that is used by the cleint([13]).

After the lease is recorded, run the command Get-DhcpServerv4Lease -ScopeId <scope> to confirm clients are receiving addresses. The cmdlet also has a means of identifying bad leases with -BadLeases and -AllLeases to specify all IPv4 address leases([14]). The logs of the server and leases are stored in an Event Viewer and it tracks major DHCP service events, but individual lease events go to seperate DHCP log files. By default, they are in the directory `C:\Windows\system32\dhcp` with one file per weekday([15]).

3. Conclusion

3.1 Summary

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3.2 Key Findings

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3.3 Future Work

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3.4 Final Remarks

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A. Appendix

A.1 Github Repository